

Accumulated FAI Events as Collective Evolution Mechanism

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Date: May 14, 2026

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This work derives from and formalizes the inter-Self coordination architecture introduced in “Inter-Self Coordination via Shared Substrate / Full Aspect Integration” (Li, April 2026), the third paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026) and “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026).

Abstract

This note formalizes D1.27, the twenty-seventh Phase D1 sub-commitment in the Series D derivation of Paper 3’s six architectural claims. D1.27 derives from Paper 3 Claim 6 (D0.06), which establishes population-scale collective evolution as the architectural object that accumulated FAI events across many CKS-governed Selves compose into at population register. D1.27 isolates the specific sub-commitment concerning the mechanism by which that evolution proceeds: two dynamics — pattern propagation and specialization preservation — operating in governed interplay to produce a convergence-with-diversity outcome at population scope. The note states both dynamics with architectural precision, explains why this constitutes governed collective evolution rather than emergent coordination, identifies the inheritance relationship to Paper 2’s three intra-Self evolution mechanisms, and states four failure modes the sub-commitment defends against. An operational test appropriate to the calibrated-humility register is provided. The claim throughout is one of architectural possibility, not outcome prediction: the architecture is designed such that accumulated FAI events CAN constitute a collective evolution mechanism with the convergence-with-diversity property. Whether specific deployments achieve this possibility depends on empirical validation, organizational dynamics, and governance practice.

1. Why this sub-commitment requires explicit statement

Paper 3 Claim 6 asserts that when accumulated FAI events across many CKS-governed Selves operate under joint authority across population-level governance perimeters, the cumulative coordination structure constitutes population-scale collective evolution as an architectural object. That assertion is the thesis of Paper 3’s §9. But the assertion compresses several distinct sub-commitments, each of which requires separate formalization to be defensible against misreading and to serve as prior art

against independent claims.

D1.27 concerns the mechanism sub-commitment: what does collective evolution across a network of CKS-governed Selves look like architecturally, and what makes it collective evolution rather than something else? The question is non-trivial because “collective evolution” is a term that could describe two quite different things: an emergent phenomenon in which patterns propagate across a network without any single governing authority orchestrating them, or a governed process in which every propagation step requires authorization at each participating governance perimeter. These two readings carry radically different implications for accountability, traceability, and sovereignty. D1.27 formalizes the second reading and defends it against collapse into the first.

The sub-commitment also surfaces a structural possibility that the architecture enables but does not automatically produce: a network of CKS-governed Selves can achieve convergence where convergence is valuable and diversity where diversity is valuable, without mandating either. This convergence-with-diversity possibility is the distinctive outcome the architecture is designed to enable. Stating it precisely, in the calibrated-humility register appropriate to an architectural claim, is the purpose of this note.

2. Statement of D1.27

D1.27 — Accumulated FAI events as collective evolution mechanism: When FAI events accumulate across a network of CKS-governed Selves operating in diverse domains, the architecture makes the following possible: the accumulated events constitute a collective evolution mechanism at population scope, operating through two complementary dynamics. Dynamic 1 (pattern propagation) enables effective coordination patterns to spread across the network wherever receiving Selves’ governance configurations authorize absorption. Dynamic 2 (specialization preservation) enables distinctive governance architectures to persist wherever Selves’ governance configurations protect them. The interplay of these two dynamics produces a convergence-with-diversity outcome at population scope — common ground where it is valuable, diversity where it is valuable — as an architectural possibility grounded in home governance authority at every participating Self.

What distinguishes this from emergent coordination is that every propagation step is individually governed: no pattern reaches a new Self without home governance authorization at that Self; no specialization is erased without home governance decision at that Self. The accumulation of individually-governed events is what constitutes the collective evolution mechanism. There is no network-level controller; there is no automatic propagation pipeline. The mechanism operates through the same FAI event structure Paper 3 Claim 4 establishes (D0.04), with the evolution-feed loci (D1.17–D1.20) providing the per-event handoff from coordination into home evolution machinery.

3. Dynamic 1: Pattern propagation

The first dynamic concerns how effective coordination patterns can spread across a network of participating Selves. The word “can” carries the full weight of the calibrated-humility register:

propagation is not guaranteed; it is possible wherever governance configurations enable it.

The mechanism operates as follows. When a FAI event concludes and the evolution-feed loci process its outputs (D1.17, the action-feedback locus; D1.18, the DNA evolution locus), each participating Self receives, as substrate content, information about the patterns the event employed. Orchestration rules that proved effective, DNA specifications that proved robust, action-feedback insights that produced consistent improvements — these are available at the evolution-feed boundary for absorption into home evolution machinery. Absorption does not happen automatically. Each receiving Self's governance configuration (D1.18) determines whether the incoming pattern enters the DNA mutation queue, the directed selection filter, or is set aside without further processing. The authority over what to absorb belongs to home governance at each receiving Self.

As FAI events accumulate across diverse domains — engineering teams coordinating with research teams, cross-functional project Selves coordinating with departmental Selves, inter-organizational collaborations — patterns that prove effective across heterogeneous contexts begin to recur in FAI event records. Patterns that are robust in many distinct governance environments show up repeatedly. Governance configurations that enable absorption of such patterns allow the network to develop a shared repertoire — not through central prescription, but through the accumulation of individually-governed absorptions. Where many Selves authorize absorption of the same pattern, that pattern propagates. Where Selves do not authorize absorption, the pattern does not reach them.

This is propagation as governed selection operating at network scope: each individual absorption decision is a governance action; the statistical aggregate of many such decisions is the propagation dynamic.

4. Dynamic 2: Specialization preservation

The second dynamic concerns how distinctive governance architectures are maintained rather than dissolved by participation in FAI events. A Self operating in a domain with specialized requirements — domain-specific orchestration rules calibrated to regulatory environments, content domains reflecting unique organizational knowledge, operational approaches developed over many governance cycles — retains those specializations not by refusing to participate in FAI events but by governing what it absorbs from them.

The mechanism operates through two governance levers. First, participation configuration (D1.26): a Self's governance authority can configure the scope and terms of participation in FAI events, restricting what aspects of the home substrate are exposed to the shared event substrate and what aspects remain within home perimeter. Second, absorption configuration (D1.18): even where the evolution-feed loci deliver incoming patterns to the home perimeter, home governance determines what enters the evolution machinery and what is declined. A Self can participate fully in FAI events for coordination purposes while absorbing very little from those events into its own evolution machinery — maintaining specialization while enabling coordination.

Specialization preservation is thus not an architectural constraint imposed on the network; it is a governance capability available to each participating Self. Selves choose, through governance configuration, how much to converge with network patterns and how much to maintain distinctive approaches. The architecture preserves this choice; it does not prescribe the answer.

5. The convergence-with-diversity outcome as architectural possibility

The interplay of Dynamic 1 and Dynamic 2 produces a structural possibility at population scope: convergence-with-diversity. Common ground where it is valuable — shared coordination patterns, stable exchange vocabularies, orchestration approaches that have proven robust across diverse contexts — can emerge from many Selves independently authorizing absorption of the same effective patterns. Diversity where it is valuable — specialized governance architectures adapted to specific domains, unique operational knowledge preserved within home perimeters — persists where Selves govern against dissolution.

This outcome is an architectural possibility, not a guaranteed result. The claim is not that a network of CKS-governed Selves will achieve convergence-with-diversity, or that this balance is always reachable. The claim is that the architecture is designed such that this outcome CAN emerge from the accumulated FAI events of many participating Selves, without requiring central coordination of the balance and without mandating either convergence or diversity.

Two failure modes of the balance — all-convergence and all-diversity — are architecturally distinguishable. All-convergence would occur if governance configurations at most Selves authorized absorption of most incoming patterns without filtering, eroding specializations faster than governance decisions can protect them. All-diversity would occur if governance configurations at most Selves declined most incoming patterns, leaving each Self isolated in its evolution trajectory without learning from the broader network. Neither failure mode is mandated by the architecture; both are possible outcomes of governance choices made at the level of individual Selves. The architecture's contribution is to preserve the governance levers that allow Selves to navigate between these failures — and to do so at each participating Self independently rather than through population-level control.

6. Governed collective evolution versus emergent coordination

The distinction between governed collective evolution and emergent coordination is the diagnostic feature of D1.27. It is worth stating with precision because the two can produce superficially similar network-level outcomes — patterns becoming common across many agents, diversity persisting in specialized contexts — while differing fundamentally in their accountability structure.

In emergent coordination approaches, patterns propagate through mechanisms outside governance: statistical regularities in model behavior, shared training data producing similar instincts, runtime communication that produces alignment without explicit governance action. The accountability question — who authorized this pattern's spread? — may not have a traceable answer because propagation did not pass through governance. This produces a structural gap between what the network

does collectively and what any participating governance authority decided.

In the architecture D1.27 formalizes, every propagation step passes through home governance authorization. A pattern reaches a new Self only because that Self's governance configuration enabled absorption. A specialization persists at a Self only because that Self's governance configuration protected it. The sequence of governance actions that produced any observed network-level pattern is, in principle, traceable through the governance records of participating Selves. This traceability is not incidental; it is the architectural property that makes collective evolution compatible with the human governance commitments Paper 1 establishes and Paper 3 extends to population scope.

Three properties follow from this governance structure. First, accountability is locatable: for any observed pattern propagation, there exists a set of home governance authorization records at receiving Selves from which the propagation can be reconstructed. Second, reversibility is available: a Self can revise its governance configuration to decline future absorption of a pattern it has previously absorbed, or to re-enable absorption of a pattern it has declined. Third, no central coordinator is required: the collective evolution mechanism operates through the home governance decisions of many individual Selves, not through a population-level authority that decides what propagates.

7. Inheritance from Paper 2: a fourth evolution mechanism at network scope

Paper 2 establishes three evolution mechanisms within a CKS-governed Self: instinct-layer mutation, DNA-layer directed selection, and action-feedback. These three mechanisms operate in productive tension (Paper 2 §7), with each mechanism's governance shape co-determined with its structural properties (Paper 2 §8). All three operate within the perimeter of a single Self.

D1.27 identifies a fourth evolution mechanism that operates at network scope: the collective dynamics produced by accumulated FAI events across many Selves. This fourth mechanism is not a new architectural primitive added at population scope; it emerges from the operation of the existing FAI event structure (Paper 3 Claim 2, D0.02) and the evolution-feed mechanism (Paper 3 Claim 4, D0.04) applied across many events. The action-feedback locus (D1.17) and the DNA evolution locus (D1.18) are the per-event handoff points at which each individual FAI event connects to home evolution machinery. When many such events accumulate, and many Selves' governance configurations authorize absorption through those loci, the aggregate constitutes the network-scope evolution mechanism D1.27 describes.

The fourth mechanism inherits the governance structure of the first three. As governance shapes are co-determined with mechanism shape at intra-Self scope (Paper 2 §8), governance configurations at the FAI event scope — participation configuration, absorption configuration, specialization protection — are co-determined with the network-scope collective evolution mechanism's behavior. No new governance infrastructure is required at population scope to govern the fourth mechanism; home governance at each participating Self, exercised through the governance levers available at the FAI hand-off boundary, is sufficient.

8. Four failure modes the sub-commitment defends against

Failure mode 1 — Ungoverned emergent coordination. The claim that patterns propagate across the network without home governance authorization at each receiving Self — that the network-level coordination dynamics operate outside the governance of any participating Self. D1.27 defends against this by establishing that each propagation step requires governance authorization at the receiving Self, through the absorption configuration available at the DNA evolution locus (D1.18). No pattern reaches a Self without that Self's governance enabling it.

Failure mode 2 — Central coordinator required. The claim that collective evolution across a network of CKS-governed Selves requires a population-level controller — an authority that decides what propagates, what specializations survive, and how to balance convergence with diversity. D1.27 defends against this by establishing that the collective evolution mechanism operates through the accumulation of home governance decisions at individual Selves. The population-level outcome — convergence-with-diversity — is the aggregate of many individually-governed events, not the output of a central decision authority.

Failure mode 3 — Mandatory convergence. The claim that the architecture prescribes that all Selves participating in FAI events must converge toward common patterns — that specializations cannot be preserved, and diversity is an architectural failure. D1.27 defends against this by establishing Dynamic 2 as a co-equal architectural property: specialization preservation is available to every participating Self through governance configuration, and the architecture is designed to enable it.

Failure mode 4 — Non-evolutionary FAI. The claim that FAI events are coordination events only — that they produce no evolution effects and connect to no home evolution machinery. D1.27 defends against this by grounding the collective evolution mechanism in the four-locus evolution-feed established in Paper 3 Claim 4 (D0.04): every FAI event connects to home evolution machinery through the dissolution hand-off loci, producing evolution effects at every participating Self wherever governance configurations enable absorption.

9. Operational test

An operational test for D1.27 must be calibrated to the architectural possibility register: it does not test whether convergence-with-diversity has been achieved, but whether the architecture makes it traceable when it does occur. The test is as follows.

Across a network of FAI-participating Selves, an observer selects a DNA pattern — an orchestration rule or content-domain specification — that appears in the governance records of multiple Selves that have participated in common FAI events. The test asks: can the observer trace this pattern's propagation through the FAI event records?

Specifically: for each receiving Self at which the pattern appears in post-event governance records and did not appear in pre-event records, can the observer locate (a) the FAI event through which the pattern was available at the home perimeter, (b) the home governance authorization — the governance

configuration at the DNA evolution locus — that enabled absorption, and (c) the evolution record showing the pattern’s entry into the home DNA layer?

And for each Self that participated in the same FAI events but did not absorb the pattern, can the observer identify the governance configuration that prevented absorption — confirming that non-propagation was itself a governance outcome rather than an architectural gap?

If both parts of the test are satisfiable — propagation traceable through governance authorizations, non-propagation traceable through governance configurations — the architecture instantiates D1.27’s governed collective evolution commitment. If pattern appearances at receiving Selves cannot be traced to governance authorizations, the architecture has failed the sub-commitment and instantiates Failure mode 1 (ungoverned emergent coordination) instead.

The test does not require that propagation has occurred; it requires that, wherever it has occurred, the governance record is present. This is the appropriate test for an architectural possibility claim: the architecture is tested for whether it preserves the governance conditions under which convergence-with-diversity is achievable, not for whether a specific deployment has achieved it.

10. Conclusion

D1.27 formalizes the mechanism sub-commitment within Paper 3 Claim 6: accumulated FAI events across a network of CKS-governed Selves constitute a collective evolution mechanism at population scope, operating through pattern propagation where governance enables it and specialization preservation where governance protects it. The convergence-with-diversity outcome this mechanism makes possible — common ground where it is valuable, diversity where it is valuable — is an architectural possibility, not an outcome prediction.

The sub-commitment’s defining architectural property is that every step of the collective evolution mechanism is governed. Pattern propagation requires home governance authorization at each receiving Self; specialization preservation requires governance configuration at each protecting Self; the accumulation of FAI events is traceable through each participating Self’s governance records. This governance structure is what distinguishes the architecture from emergent AI coordination approaches and from approaches that require a central population-level controller.

The inheritance from Paper 2 is precise: Paper 2’s three intra-Self evolution mechanisms compose at network scope into a fourth mechanism operating through the FAI event structure Paper 3 establishes. No new architectural primitives are required at population scope; the collective evolution mechanism operates through the per-event governance machinery the preceding claims establish.

Empirical validation, organizational dynamics, and governance practice will determine whether specific deployments achieve the convergence-with-diversity possibility. The architecture’s contribution is to make that possibility available through the governance structures each participating Self exercises over its own evolution — and to make the achievement traceable when it occurs.

Source papers

Li, W. (2026a). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026b). *The Instinct/Reasoning Separation Outside the Model*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026c). *Inter-Self Coordination via Shared Substrate / Full Aspect Integration*. April 2026. ORCID: 0009-0004-8065-3235.

How to cite this note

Li, W. (2026). *Accumulated FAI Events as Collective Evolution Mechanism*. Derivation Note #492 (D1.27), CKS Theory Series D. May 14, 2026. ORCID: 0009-0004-8065-3235.